AUGMENTED DECISION

Statistical learning and classification

Decision trees using weak learners

Lecture 3 / Three hours

Theory before the break. Worked examples and a group decision case after the break.

A weak classifier can have a small advantage

A base classifier $h_t(x) \in \{0, 1\}$ is one rule we may combine with others. For an evaluation distribution Q, define its error and edge:

$$e_Q(h_t) = Q\{h_t(X) \neq Y\}, \qquad \gamma_Q(h_t) = \tfrac{1}{2} - e_Q(h_t).$$

Example: 60 correct predictions out of 100 give an empirical error of 0.40 and an empirical edge of 0.10.

- Call it weak for Q when $\gamma_Q(h_t) > 0$: it beats a fair random guess.
- A simple rule need not have an edge. Its performance can change when the data weights change.

On imbalanced data, also compare with the majority-class baseline. An observed edge does not establish a population guarantee.

Different mistakes can make a useful vote

Case	Y	h_1	h_2	h_3
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	1	1
5	1	1	0	1
6	1	1	1	0

Each rule makes two errors. The majority vote makes none on these six cases.

More generally, use weights $a_t \geq 0$ with $\sum_t a_t > 0$:

$$v(x) = \frac{\sum_t a_t h_t(x)}{\sum_t a_t},$$

$$h_{\text{vote}}(x) = \mathbf{1}\{v(x) \geq \tfrac{1}{2}\}.$$

Repeating the same rule three times leaves its mistakes unchanged.

Boosting gives more weight to mistakes

A first rule misclassifies cases 1 and 4 in the six-case table.

Cases	Initial weight per case	Next weight per case
1 and 4: incorrect	1/6	1/4
2, 3, 5, 6: correct	1/6	1/8

These are the normalized AdaBoost weights after an error of 1/3. The next rule searches for a small *weighted* error.

A rule wrong only on cases 2 and 3 has weighted error $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.

Boosting learns successive rules and combines their votes. Noisy labels may receive large weights, so choose the number of rounds with validation data.

A decision stump asks one question

Let x_1 be the number of minutes a parcel leaves the warehouse late.

$$b(x) = \mathbf{1}\{x_1 \geq 30\}, \qquad h(x) = \begin{cases} 0, & b(x) = 0, \\ 1, & b(x) = 1. \end{cases}$$

Below 30 minutes

20 historical parcels, 4 delayed.

Estimated delay probability: $4/20 = 0.20$.

Majority label: 0.

At least 30 minutes

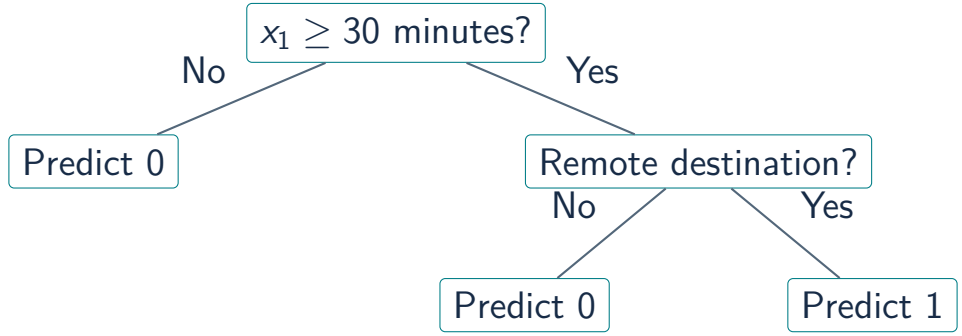
20 historical parcels, 14 delayed.

Estimated delay probability: $14/20 = 0.70$.

Majority label: 1.

A stump is a depth-one tree. Its split is a binary test; its two leaf labels complete the classifier.

A tree makes the next question conditional



A parcel that leaves 45 minutes late follows the right branch. Its destination determines the final leaf.

A vote combines outputs from several rules. A tree combines tests along a path.

The displayed tree is an illustrative rule, separate from the case-study data.

A split should create more informative groups

A node contains 10 examples: 5 positives and 5 negatives. For a positive fraction q, use Gini impurity $G(q) = 2q(1-q)$.

Group	Positive / total	q	$G(q)$
Parent	5/10	0.5	0.50
Left child	1/5	0.2	0.32
Right child	4/5	0.8	0.32

$$\text{weighted child impurity} = \tfrac{5}{10}(0.32) + \tfrac{5}{10}(0.32) = 0.32.$$

The decrease is $0.50 - 0.32 = 0.18$. Majority-label errors fall from 5 to 2.

Weight by child size. A split with 50% positives in each child would give zero decrease.

A useful tree may need two questions

x_1	x_2	Y
0	0	0
0	1	1
1	0	1
1	1	0

Four equally weighted examples.

A single split on x_1 or x_2 leaves a 50–50 mix in each child.

A depth-two tree can first ask about x_1, then about x_2 in both branches. Each leaf contains one example.

A greedy algorithm that insists on an immediate positive gain can stop too early.

Greater tree depth can capture interactions. Perfect fit to four training cases does not establish good generalization.

Validation helps choose how far to grow

Candidate tree	Training errors / 100	Validation errors / 100
Depth 1	24	27
Depth 2	12	18
Depth 6	0	25

For these hypothetical candidates, choose depth 2 if the target is classification accuracy.

- Grow or prune using training and validation data.
- Evaluate the frozen choice on an untouched test sample.
- If decisions have unequal costs, select using the relevant validation loss.

A lower training error alone cannot establish a lower population risk. Small leaves also make probability estimates unstable.

Case study: which parcels should be expedited?

You manage a fictional parcel service. Today, at most **two of six parcels** can use the express service.

- $Y = 1$ means a parcel would arrive late under standard service.
- Standard service costs 60 units if late and 0 otherwise.
- Express service costs 15 units and, for this exercise, guarantees on-time delivery.

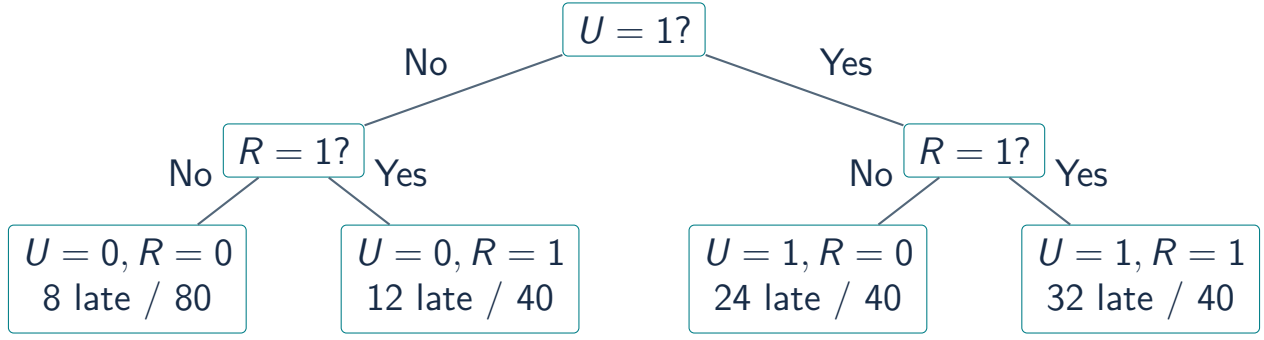
Historical training and validation parcels all used standard service. Their observed labels therefore measure standard-service delay.

Your decision must combine the tree's estimated probability, the cost, and the two-parcel capacity.

All data are synthetic. Express effectiveness is a supplied assumption, not a causal conclusion learned from the tree.

Two candidate trees fitted to 200 parcels

Features: $U = 1$ means late warehouse handover. $R = 1$ means a remote destination.



T_2: the displayed depth-two tree. Each leaf estimates $\hat{\eta}(x)$ by its late fraction.

T_1: stop after the U split, pooling the two leaves on each side.

Class prediction and dispatch use different losses

Let $a = 0$ mean standard service and $a = 1$ mean express service.

Loss $L(a, Y)$	$Y = 0$	$Y = 1$
Standard ($a = 0$)	0	60
Express ($a = 1$)	15	15

$$\hat{\rho}(0 \mid x) = 60\hat{\eta}(x), \qquad \hat{\rho}(1 \mid x) = 15.$$

For an unconstrained decision, compare these two estimated costs. For the six-parcel batch, respect the capacity as well.

Classification uses the 0–1 loss and threshold 1/2. Dispatch uses L above. The worksheet asks you to derive its threshold.

Group task: recommend a dispatch plan

Work from the student worksheet. Submit one recommendation per group.

1. compute the leaf probabilities for T_1 and T_2.
2. derive the cost threshold and the actions in each leaf.
3. compare the two policies on the 100 validation parcels.
4. choose at most two parcels from today's batch and calculate its estimated total cost.

Give the selected tree, parcel IDs, estimated cost, and one assumption your recommendation depends on.

Validation compares policies without a capacity limit. Capacity applies to today's batch. Keep training leaf estimates fixed during the comparison.

The tree informs a decision that still needs judgment

- Leaf proportions estimate standard-service delay probabilities. Small samples and changing routes can make them unreliable.
- The loss table converts probabilities into preferred actions. The capacity changes which preferred actions are feasible.
- The express guarantee is an external assumption. Different effectiveness would change the savings and threshold.
- Freeze the complete policy, including the capacity rule, before evaluating it on fresh batches.

Recommendation: use T_2 and expedite D and F, with estimated total cost 108 under the stated assumptions.

Exit question: what additional evidence would you request before using this policy in a real dispatch operation?

Appendix A: the reweighting arithmetic

For this calculation only, write $s_i = 2Y_i - 1$ and $g(x) = 2h(x) - 1$. The first rule has weighted error $\epsilon = 1/3$, hence coefficient

$$c = \tfrac{1}{2} \log \frac{1 - \epsilon}{\epsilon} = \tfrac{1}{2} \log 2.$$

Multiply each weight by $\exp(-c\, s_i g(X_i))$ and normalize.

- Incorrect cases receive factor $\sqrt{2}$.
- Correct cases receive factor $1/\sqrt{2}$.

Starting from $1/6$, normalization gives $1/4$ for each of two incorrect cases and $1/8$ for each of four correct cases.

This is the binary AdaBoost update of Freund and Schapire. The main session uses its numerical effect only.

Appendix B: notation used in the case

Symbol	Meaning
$X = (U, R)$	Features observed before choosing service
$Y \in \{0, 1\}$	Delay outcome under standard service
$\eta(x)$	Unknown standard-service delay probability
$\widehat{\eta}(x)$	Training late fraction in the reached leaf
$h(x)$	Predicted class, using threshold $1/2$
$a \in \{0, 1\}$	Operational action: standard or express
$L(a, Y)$	Operational cost from the supplied loss table
$\widehat{\rho}(a \mid x)$	Expected cost with η replaced by $\widehat{\eta}$
T_1, T_2	Stump and depth-two candidate tree

The first half uses R_P for population risk. The capital R inside (U, R) denotes the remote-destination indicator only.

Appendix C: reading and teaching materials

Weak classifiers and boosting

Y. Freund and R. E. Schapire (1999), *A Short Introduction to Boosting*.

Author-hosted introduction and binary AdaBoost algorithm

[SL] Learning theory and decision trees

S. Shalev-Shwartz and S. Ben-David (2014), *Understanding Machine Learning: From Theory to Algorithms*, Chapters 4, 10, and 18.

Book website

Companion materials

Student worksheet: two pages with all data and response space.

Instructor solution: calculations, timing, marking guide, and an optional sensitivity exercise.

All numerical examples and the parcel-service data in the revised second half are synthetic teaching examples.